Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



## Team Details

a. Team name: **BreederNear AI**
b. Team leader name: **Sudarsan N**
c. Team member: **Shreya Azad**
d. Problem Statement: **Retail & Commerce: trusted, legal breeder-to-buyer pet commerce in India (BreederNear AI)**

# Brief about the idea

## “The breeder is 50 metres away. The customer never finds them.”

### Why we are building this

Our founder Sudarsan N spent five years as a bird breeder, pet-shop owner and broker in Coimbatore. He started with a ₹750 setup and earned over ₹4 lakh while knowing only about 100 people. Every week he watched buyers walk past a breeder on their own street into a shop, pay more for a weaker bird, or lose money to a stranger in a WhatsApp group.

### Buyers lose money and trust

- Can’t find the breeder next door, so they pay shop prices
- Can’t tell a real seller from a scam: “full advance, courier only”, stolen photos
- Don’t know which birds are protected or that dog breeders must register[5]
- Alone after buying: wrong cage, wrong food, late to the vet

### Breeders lose time and reach

- Sell only inside their own small circle and district groups
- Hours of calls every day, the same 20 questions, the same post retyped in every group
- Good breeders look exactly like bad ones: no way to prove it

### Animals and the law lose most

- 300+ of India’s ~1,200 bird species are found in illegal trade[3]
- Munias are dyed to disguise them[4]; dyed chicks were still on sale in an unlicensed Bengaluru shop in June 2026[6]
- Sick animals are sold because nobody checks before the sale

**Still unsolved in 2026:** OLX stopped live-animal listings[2], big pet platforms sell food and accessories, so India’s ~39 million-pet market[1] still buys animals through WhatsApp, Facebook and Instagram groups with no checks at all.

Founder figures are his own experience · [1] Euromonitor 2025 · [2] OLX India: items not allowed · [3] WWF-India / TRAFFIC · [4] Deccan Herald, Aug 2023 · [5] PCA (Dog Breeding and Marketing) Rules 2017 · [6] PETA India, 25 Jun 2026

# From pain to product: the whole journey, end to end

| Step | Today, in groups and shops | With BreederNear AI | Built and measured |
|---|---|---|---|
| Find | Walk into the nearest shop; the breeder next door is invisible | Pets near you, trusted first, a match % for your home, a fair-price bar | Match % calculated in code; 16 sample breeders, 36 sample pets |
| List | Breeder retypes the same post in every group | Photos + a Tamil or English message, typed or as a voice note, become a full listing | 17–27 s live: upload, AI fill, trust check |
| Trust | No checks: stolen photos, scam prices, protected birds | Every listing screened before it goes live; Verified breeder badge | **8/8 protected species blocked, 0/37 false alarms** |
| Check | Pay a stranger on social media in full, in advance | Paste any outside post: “Is this post safe?”; share the warning back to the group; report protected animals | 5–7 s per post; nothing stored |
| Talk | Endless calls and the same answers | AI drafts the buyer’s first message and the breeder’s reply; scam shield on both | Scam scan on every enquiry and reply |
| Buy | “Courier only”, no handover | Book a visit, see the pet, pay in person, tap “I bought this pet” | No payments in the app, by design |
| After | New owner on their own | Starter kit sized for welfare, 14-day plan, daily log, growth, symptom urgency | **Never a medicine or dose (evals E11, E19)** |
| Grow | Breeder has no data | Dashboard: views, enquiries, visits and sales, week on week | 8-week charts and trend cards |

**One tool layer, two ways in:** every step works by tapping or by asking seven Gemini agents in English or Tamil. Live now: 69/69 agent eval runs, 339 unit tests, 0 accessibility problems.

# Opportunities

## How is it different?

**WhatsApp / Facebook groups:** reach, but no checks, no records, no recourse

**Classifieds (OLX):** stopped live-animal listings

**Pet super-apps:** food, accessories and vets, not the animal

**Kennel websites:** one seller vouching for itself

**No platform we found checks every listing for wildlife law, registration, scam prices and reused photos before it goes live, and then stays with the buyer after the sale.**

## How does it solve the problem?

**Invisible breeders** → listed in a minute from photos and a Tamil message

**Scams** → price, wording and reused-photo checks; outside posts checked too

**Illegal birds** → blocked in code from the photo alone

**Unproven breeders** → Verified badge: phone, facility photos, registration

**Lost hours** → AI drafts replies; visits booked in the app

**Alone after buying** → starter kit, care plan, daily log, vet urgency

## USP

**Trust built in, not bolted on.**

- Rules in code decide; Gemini reads and explains. A prompt injection still ends in CAUTION (eval E12).
- Protected species blocked from the photo alone: 8/8 in testing, 0/37 false alarms.
- Works where the trade already happens: Tamil and English messages, posts from groups.
- Founder–market fit: designed from five years inside this exact trade.

# List of features offered by the solution

## For buyers (customer app)

- Pets near me with a match % for your home, trusted first, fair-price bar
- Listing page: trust checks, health notes, questions to ask
- AI-drafted first message; scam warnings on replies
- Book a visit; “I bought this pet” after it (paid in person)
- My pets: daily log with AI feedback, growth, health calendar
- Symptom checker: how urgently to see a vet (never medicines)
- “Is this post safe?” with share and report; starter kit; Local Breeders

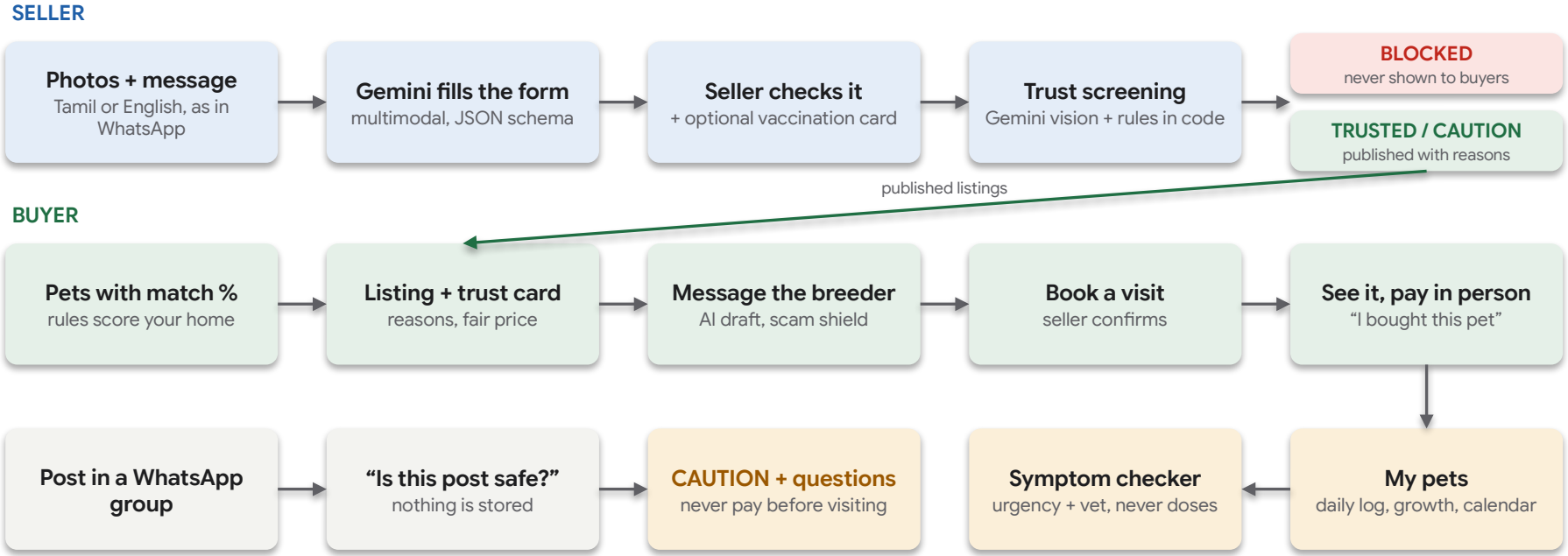
## For sellers (seller app)

- “Fill with AI”: photos + a Tamil/English message or voice note become a listing
- Publish with trust check: TRUSTED, CAUTION or BLOCKED, with reasons
- Dashboard charts: views, enquiries, visits, sales, funnel
- Verified breeder badge: phone code, facility photos read by Gemini, dog-breeder certificate (no ID documents)
- Visit requests to confirm; AI-drafted replies to enquiries
- Vaccination card read into the trust check

## Everywhere

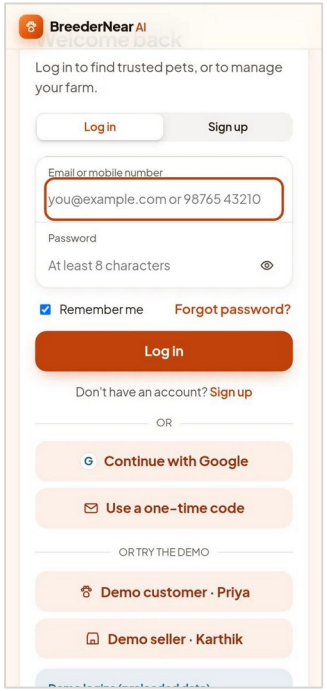
**Seven Gemini agents** in English or Tamil use the same tools as the screens. **Sign-in:** password, one-time code or Google (Firebase Authentication). **Safety:** roles checked in the API and the AI tools, rate limits, every AI reply checked for medicine doses. **Accessible:** WCAG 2.2 AA, phone to desktop.

# Process flow diagram or Use-case diagram

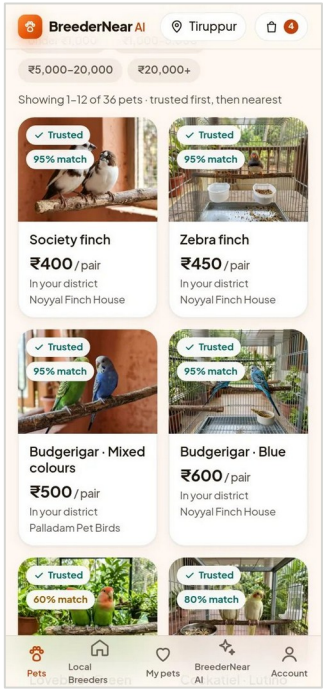



Every step works by tapping, or by asking BreederNear AI in chat: the same tools, the same checks. Sellers also earn a Verified breeder badge.

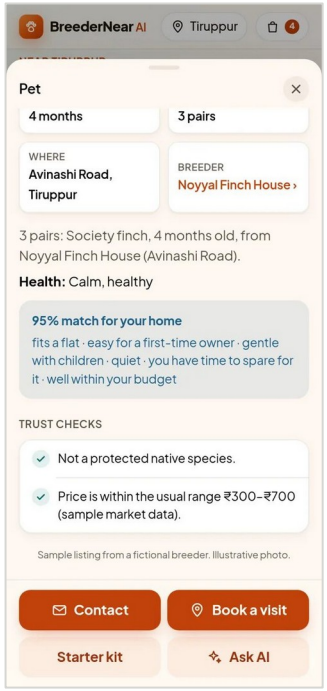
# Wireframes/Mock diagrams of the proposed solution (optional)



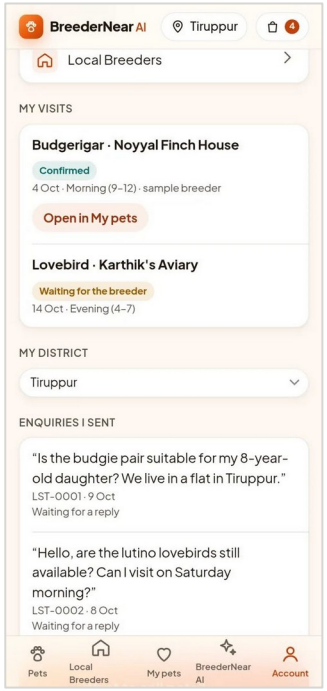

Sign in or one-tap demo



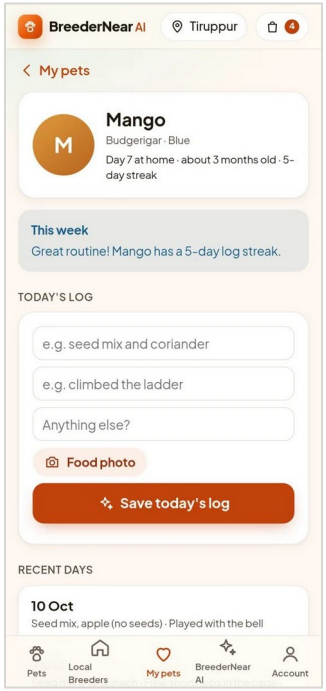

Pets with match %




Listing + trust checks

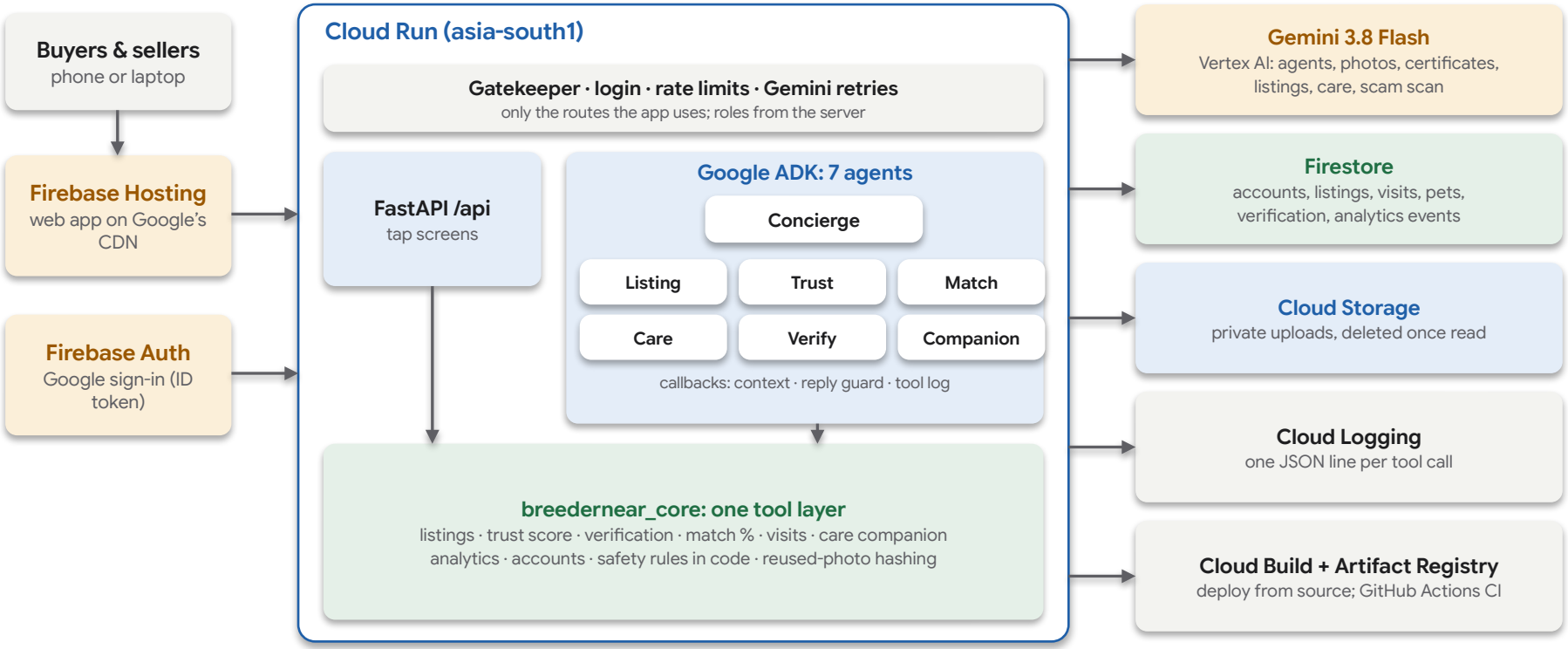



Visits, then "I bought it"




My pets companion

# Architecture diagram of the proposed solution



# Technologies to be used in the solution

| Google technology | How BreederNear AI uses it |
|---|---|
| Gemini 3.8 Flash (Vertex AI) | Seven ADK agents; listings from photos, Tamil/English text and voice notes; photo, facility and certificate reading; care plans, daily-log feedback, symptom urgency; message drafts and scam scan. JSON schemas. |
| Google ADK (Python) | Concierge + listing, trust, match, care, verification and companion agents; tools; callbacks on every agent; streamed replies |
| Cloud Run | API and agents; login gatekeeper; rate limits; Gemini retries with backoff |
| Firebase Hosting | The web app on Google's CDN; calls Cloud Run directly (CORS) so AI replies stream |
| Firebase Authentication | Continue with Google; the server verifies the Firebase ID token |
| Firestore | Accounts, listings, enquiries, visits, sales, pets and logs, verification, analytics |
| Cloud Storage, Cloud Build, Artifact Registry, Cloud Logging | Private uploads; deploy from source; tool-call logs |
| gemini-3.1-flash-image | Illustrative photos for the fictional sample listings (labelled in the app) |

## Also used

- Python 3.12, FastAPI, Pydantic
- Pillow: our own perceptual hash (dHash) for reused photos
- Plain HTML, CSS, JavaScript and SVG charts: no build step
- pytest, Playwright + axe-core, GitHub Actions
- Firebase CLI (deploy only)

# Google Cloud AI map: which AI does each job, and how well

| Google Cloud AI | What it does in BreederNear AI | Measured |
|---|---|---|
| Gemini vision | Listing photos: species, protected birds, dye, blur, watermarks, stock photos. Facility photos, dog-breeder certificates, vaccination cards, food photos | **8/8 protected species blocked; 0/37 false alarms; 202/202 photo checks** |
| Gemini audio | A breeder's Tamil or English voice note becomes a full listing, with a transcript; no separate speech-to-text step | **Tamil voice note → correct listing (3/3 eval runs, E23)** |
| Gemini, multilingual | Romanised Tamil and English sale messages; a buyer's own words become a home profile | Tamil message → same listing (E02); profile → match % (E21) |
| Gemini structured output | 12 task types, each with a JSON schema. Gemini reads and writes; code decides trust, badges, match % and urgency | Prompt injection still ends in CAUTION (E12) |
| Google ADK | 7 agents (a concierge and 6 specialists) and 29 tools, the same tools the tap screens use; streamed replies | **69/69 agent eval runs passed (23 cases × 3 runs)** |
| ADK callbacks | Context before every model call; every reply checked for medicine names and doses, even while streaming; every tool call logged | No drug names or doses (E11, E19) |
| Vertex AI | Gemini 3.8 Flash on the global endpoint, retries with backoff on 408, 429 and 5xx | Listing in 17–27 s; post check 5–7 s |
| Firebase Auth, Cloud Run, Firestore, Storage | Google sign-in; the API and agents; all data; private uploads | Smoke test 17/17 on every release |

# Estimated implementation cost (optional)

# Business case, scale and go-to-market

**Market:** India's pet care products market is USD 9.2 B in 2026, heading to USD 13.9 B by 2034[7], with about 39 million pets[1]. The animals themselves are still sold with no checks: that is our wedge.

## Revenue (planned)

- Verified-breeder subscription: badge, dashboard insights, more listings
- Featured listings per district
- Starter-kit and accessory commerce after every sale (cart already built)
- B2B trust-screening API for classifieds and pet shops

## Built to scale

- Trust costs ≈ ₹1.3 per listing (measured) against pets priced ₹400–₹25,000
- Serverless Cloud Run + Firestore: near zero when idle, scales out per request
- Districts, species, fair prices and protected lists are data, not code: a new district is a data file
- Gemini already reads Tamil, English and mixed messages

## Go-to-market

- 1 · Pilot in Coimbatore and Tiruppur with the founder's breeder network
- 2 · Tamil Nadu, district by district; list by sending a WhatsApp message
- 3 · Other states and languages; the trust API for other marketplaces
- Buyers arrive through "Is this post safe?" links shared in the same groups

**Growth loop:** verified breeders → trusted listings → more buyers → more sales and care data → sharper fair prices and trust scores → more breeders join.

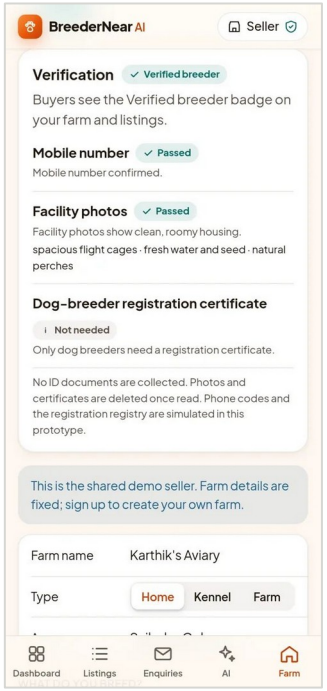
**Impact we measure, live in the app:** listings checked before going live · protected animals stopped · scam warnings raised · verified breeders. Every warning can be shared back to the group it came from, and protected species come with a way to report them.

Revenue lines and rollout are plans, not results · [1] Euromonitor 2025 · [7] IMARC Group, India Pet Care Products Market (2026) · Cost: docs/submission/05-deck-sources.md

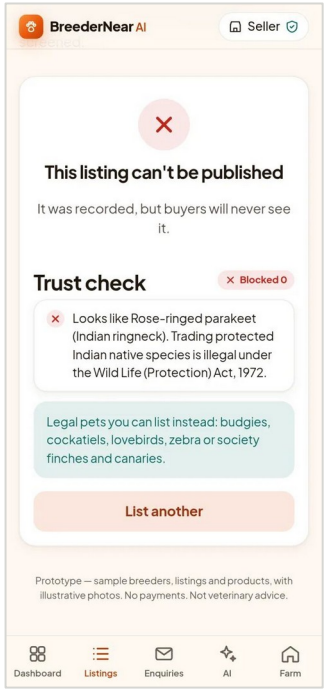
# Snapshots of the prototype



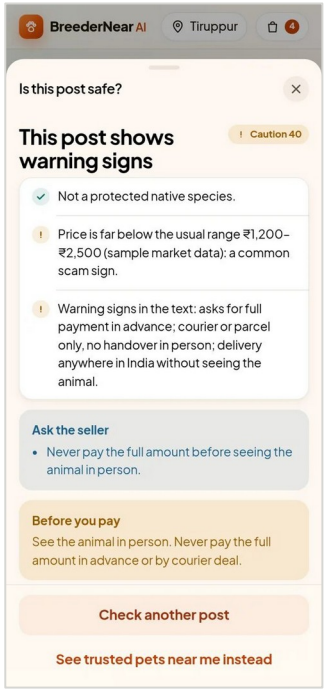

Seller analytics



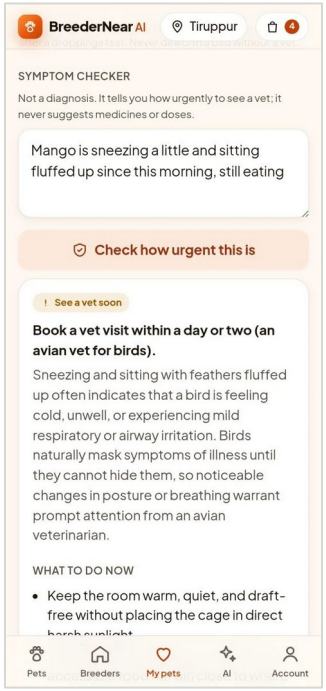

Verified breeder




Protected species: blocked




“Is this post safe?”




Symptom urgency, no doses

# Prototype Performance report/Benchmarking

| Quality (real Gemini, 10–12 Oct 2026) | Result |
|---|---|
| Agent evals: 23 cases × 3 runs (listing, Tamil, blocking, scams, reused photos, prompt injection, no medicine advice, roles, verification, visits, care companion) | 69/69 passed |
| Protected species blocked from the photo alone | 8/8 |
| Ordinary pets wrongly flagged as protected | 0/37 |
| All photo checks: species, dye, blur, no animal, watermark, screenshot | 202/202* |
| Accessibility: axe-core, WCAG 2.2 AA, light + dark, 320–1280 px, all screens | 0 problems |
| Unit tests in CI · live smoke test | 339 · 17/17 |

| Speed (live) | Time |
|---|---|
| List a pet: upload, AI fill, publish with trust check | 17–27 s |
| Check a pasted post | 5–7 s |
| Web app load (Firebase CDN) | ≈ 0.7 s |
| Gemini: read a certificate · scan a message | 8–9 s · 2–3 s |
| Gemini: symptom urgency · daily-log feedback | 7–8 s · 4–10 s |

### How we test

Gemini isn't deterministic, so each eval runs three times and we report the count. Decisions (trust, verification, match %, urgency floor) are made by tested code.

*Test photos are AI-generated and the screening prompt was tuned on them (first run 196/202), so treat this as an upper bound · Reproduce: scripts/run_evals.sh 3 · scripts/vision_accuracy.py · pytest -m ui tests/ui · evals/RESULTS.md, evals/VISION_RESULTS.md

# Additional Details/Future Development (if any)

## Roadmap (not in the prototype)

- Pilot with real breeders in Coimbatore and Tiruppur, from the founder's network
- Real SMS and email delivery for one-time codes
- List by sending a WhatsApp message (WhatsApp Business Platform)
- Live State Animal Welfare Board and PARIVESH registry checks; the platform registrations India's pet-trade rules require
- Chat history in Agent Engine sessions, so Cloud Run can scale past one instance
- Breeder ratings from buyers' care-companion outcomes; vet and pet-pharmacy partners; more languages

## Responsible by design (already built)

- Protected species blocked in code, whatever the model says
- No medicine names or doses: checked on every AI reply, even while streaming
- Verification without ID documents or selfies; photos deleted once read
- Codes shown on screen only open new or demo accounts, never someone else's
- Symptom checker: red flags are always an emergency; never a diagnosis
- Synthetic sample data, clearly labelled; simulated registry numbers start with SIM-

## Provide links to your:

- GitHub Public Repository: https://github.com/sudarsan-22/breedernear-ai
- Demo Video Link (3 Minutes): https://drive.google.com/file/d/1R4uJceUVbF6Lt0R7gGpk-nVUlvT33xrd/view?usp=sharing
- Final Product Link: https://breedernear-ai-2026.web.app (API: https://breedernear-655711985039.asia-south1.run.app)

**Try it in 2 minutes: demo accounts with preloaded data**

**Customer:** priya.customer@example.com **Seller:** karthik.seller@example.com **Password:** demo12345

Or tap “Demo customer · Priya” / “Demo seller · Karthik” on the login screen. The README has a step-by-step walkthrough.

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# Thank you

